

Table 1: Three-seed multi-suite $\pi_{0.5}$ ASRR evaluation on LIBERO. A single shared ASRR refiner is trained from a balanced cache over all four suites and evaluated with one global residual scale, $\alpha = 4.0$. Each seed uses 2000 paired rollouts. Values report mean $\pm$ sample standard deviation across eval seeds. Base and ASRR columns report success rates; Gain is in percentage points.

Suite	Base (%)	ASRR (%)	Gain (pp)	Rescue	Harm
LIBERO-Spatial	85.0 ± 0.2	93.6 ± 0.8	$+8.6 \pm 0.9$	64.3 ± 0.6	21.3 ± 4.5
LIBERO-Object	86.8 ± 1.7	94.7 ± 0.6	$+7.9 \pm 2.1$	59.7 ± 7.2	20.3 ± 4.2
LIBERO-Goal	70.1 ± 3.9	75.2 ± 1.1	$+5.1 \pm 4.6$	58.3 ± 13.3	33.0 ± 10.5
LIBERO-10	68.8 ± 0.7	72.5 ± 0.8	$+3.7 \pm 0.2$	80.0 ± 6.1	61.3 ± 6.7
Average	77.7 ± 1.2	84.0 ± 0.8	$+6.3 \pm 1.8$	262.3 ± 24.5	136.0 ± 18.7

Table 2: Raw multi-suite $\pi_{0.5}$ ASRR results for each eval seed. Each suite uses 500 paired rollouts per seed, and the Average row aggregates 2000 paired rollouts per seed.

Seed	Suite	Episodes	Base (%)	ASRR (%)	Gain (pp)	Rescue/Harm
7	LIBERO-Spatial	500	85.2	92.8	+7.6	64 / 26
7	LIBERO-Object	500	88.0	94.0	+6.0	55 / 25
7	LIBERO-Goal	500	74.6	74.4	-0.2	43 / 44
7	LIBERO-10	500	68.0	71.6	+3.6	76 / 58
7	Average	2000	79.0	83.2	+4.3	238 / 153
17	LIBERO-Spatial	500	85.0	94.4	+9.4	64 / 17
17	LIBERO-Object	500	87.6	95.0	+7.4	56 / 19
17	LIBERO-Goal	500	68.0	76.4	+8.4	65 / 23
17	LIBERO-10	500	69.0	73.0	+4.0	77 / 57
17	Average	2000	77.4	84.7	+7.3	262 / 116
27	LIBERO-Spatial	500	84.8	93.6	+8.8	65 / 21
27	LIBERO-Object	500	84.8	95.0	+10.2	68 / 17
27	LIBERO-Goal	500	67.8	74.8	+7.0	67 / 32
27	LIBERO-10	500	69.4	73.0	+3.6	87 / 69
27	Average	2000	76.7	84.1	+7.4	287 / 139